

Math 526, F08
Quiz 8

Name: *Answer*
SSN: xxx-xx-

Instructions: There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given two vectors $\mathbf{a}_1 = (1, 1, 0)$ and $\mathbf{a}_2 = (1, 1, 1)$. Find the projection matrix $\mathbf{P}$ for the subspace in $\mathbb{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

Solution: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ $A^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \Rightarrow (A^T A)^{-1} = \frac{1}{2 \times 3 - 2 \times 2} \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix} \\ = \begin{bmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{bmatrix}$$

$$\Rightarrow \mathbf{P} = A(A^T A)^{-1} A^T \\ = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Problem 2. (10 points) Find the best fitting line $y = C + Dt$ for $b = 1, 2, 4$ at times $t = -1, 0, 2$.

(a) Write down the resulting equation $\mathbf{Ax} = \mathbf{b}$.

Solution:

$$\begin{array}{ccc} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} C \\ D \end{bmatrix} & = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \\ \downarrow & \downarrow & \downarrow \\ \mathbf{A} & \mathbf{x} & \mathbf{b} \end{array}$$

(b) Solve the normal equation for the least squares fitting, $\mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}$, to find C and D .

Solution:

$$\begin{aligned} \mathbf{A}^T \mathbf{A} &= \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix} \\ \mathbf{A}^T \mathbf{b} &= \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} 3 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} &= \begin{bmatrix} 7 \\ 7 \end{bmatrix} \\ \Rightarrow \begin{cases} C = 2 \\ D = 1 \end{cases} \end{aligned}$$

Thus the best fitting line is

$$y = 2 + t$$